

Programming Assignment 2 – Bufferbloat

CSC458 – Computer Networking Systems
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1 Questions

- 2 pts. **Why do you see a difference in webpage fetch times with small and large router buffers?**

With smaller buffers, the queue is shorter so the latency is also lower, and so there are faster webpage fetches since the RTT drops. With larger buffers, many packets can occupy the queue, which keeps throughput high but also makes packets wait longer in the queue, increasing delay. As a result, webpage fetch times go up. This can be seen from one of the results obtained:

bb-q100: average=1.815298, stddev=0.242084

bb-q20: average=0.543328, stddev=0.581217

- 2 pts. **Bufferbloat can occur in other places such as your NIC.** Use `ifconfig` or `ip link show` on your Mininet VM to identify the transmit queue length (`txqueuelen`) of an interface such as `enp0s1` and report the output. For this queue size and a draining rate of 100 Mbps, what is the maximum time a packet might wait before leaving the NIC?

The output of `ifconfig enp0s1` shows a `txqueuelen` of 1000. Assuming a packet size of 1500 bytes, the maximum time a packet might wait can be calculated as follows:

$$\text{Bits per packet} = 1500 \text{ bytes} \times 8 = 12,000 \text{ bits}$$

$$\text{Total bits in queue} = 12,000 \text{ bits/packet} \times 1000 \text{ packets} = 12,000,000 \text{ bits}$$

$$\text{Draining rate} = 100 \text{ Mbps} = 100,000,000 \text{ bits/s}$$

$$\text{Maximum queueing time} = \frac{\text{Total bits in queue}}{\text{Draining rate}} = \frac{12,000,000}{100,000,000} \text{ s} = 0.12 \text{ s} = 120 \text{ ms}$$

Therefore, a packet could wait up to 120 ms before leaving the NIC.

- 3 pts. **Analyze your plots of CWND, RTT, and queue size.**

i. **Derive or express a symbolic equation showing how RTT varies with queue size.**

Based on the plots, one can see that the RTT increases linearly with the queue size. This can be expressed as the sum of base propagation delay and queuing delay as follows:

$$RTT(Q) = RTT_0 + \frac{Q}{C}$$

Where Q represents the number of packets in the queue, and C represents the bandwidth of the bottleneck link capacity in packets/second.

ii. Explain how CWND oscillations correspond to RTT spikes.

Each time TCP additively increases the CWND, more data is injected into the network, causing the queue to fill. As the CWND grows in upward oscillations in the plot, the number of packets in the queue increases, which increases the RTT and produces corresponding spikes. When the queue reaches capacity and packet loss occurs, TCP reduces the CWND multiplicatively, and the RTT spikes fall as the CWND plot oscillates downward.

iii. Discuss how buffer size influences TCP performance and webpage fetch times.

Based on the plots for `bb-q100`, larger buffers allow TCP to achieve higher CWND sizes and higher throughput, but they also increase RTTs. This means more data can be sent, but webpage fetch times can be longer when the queue approaches full capacity. As seen in the plots for `bb-q20`, shorter buffers cause packets to be dropped earlier, leading to more frequent reductions in CWND and smaller RTTs. As a result, webpages experience lower queuing delays and faster responses, but overall throughput may be slightly reduced. This behavior is also reflected in the results mentioned in my answer for question 1.

3 pts. **Identify and describe two ways to mitigate bufferbloat.** For each, explain trade-offs and propose a Mininet experiment to evaluate quantitative effectiveness. Clearly document your experimental design, including parameter choices, measurement methodology, and justification for your conclusions, so that the TA can reproduce your results.

Method 1: Smaller buffer with Explicit Congestion Notification (ECN)

- **Trade-offs:** Smaller buffers reduce queuing delay and lower RTT, but increase the chance of packet loss under congestion. ECN provides early feedback for congestion, but requires marking or dropping packets to signal the sender.
- **Parameter choices:** Enable ECN on the bottleneck link `s0-eth2`. Compute metrics such as `avg_throughput` using the results from `iperf`.
- **Measurement methodology:** Compare mean web fetch time, standard deviation, `avg_throughput`, and RTT for `maxq=20` with ECN enabled versus `maxq=20` and `maxq=100` without ECN.
- **Justification:** Smaller buffers reduce RTT but may increase packet loss. Enabling ECN allows TCP to react to congestion earlier, marking packets instead of dropping them, which mitigates bufferbloat.

Method 2: Active Queue Management (Random Early Detection, RED)

- **Trade-offs:** RED monitors the average queue length and drops packets probabilistically before the queue is full. This prevents long queues and reduces bufferbloat, keeping RTT low. Parameter tuning is difficult and may reduce throughput.
- **Parameter choices:** Enable RED on the bottleneck link `s0-eth2`. Configure RED with `min_threshold`, `max_threshold`, `drop_probability`, and `avg_packet_size`. Compute `avg_queue_length` using `count`.
- **Measurement methodology:** Compare mean web fetch time, standard deviation, RTT, queue occupancy, and CWND sizes for RED versus non-RED.
- **Justification:** By actively monitoring the average queue length, RED allows TCP to react to congestion earlier, reducing RTT and mitigating bufferbloat.

2 Theoretical Analysis

3 pts. **Sent packets per cycle.** Consider the CWND-RTT graph. Determine what is the cycle between the minimum congestion window and the maximum congestion window. Assume that no random loss occurs. Now, by summing the packets sent over RTT, show that the total number of packets sent through a cycle is roughly equal to $N_{\text{packets}} \approx \frac{3}{8}W_{\text{max}}^2$.

i. **Determining cycle between $W_{\min}$ and $W_{\max}$**

Assume TCP congestion control follows AIMD and no random loss occurs. After a loss, the congestion window is:

$$W_{\min} = \frac{W_{\max}}{2}.$$

Assume that after every RTT, the CWND is increased by at least 1 packet, denoted by k . So,

$$W(k) = W_{\min} + k, \quad k = 0, 1, 2, \dots$$

Loss occurs when $W(k) = W_{\max}$. Let K = number of RTTs until a loss occurs. Thus:

$$W_{\max} = W_{\min} + K,$$

Substituting $W_{\min}$ with $\frac{W_{\max}}{2}$, we obtain:

$$K = W_{\max} - W_{\min} = W_{\max} - \frac{W_{\max}}{2} = \frac{W_{\max}}{2}.$$

Thus, the cycle duration between $W_{\min}$ and $W_{\max}$ is:

$$K = \frac{W_{\max}}{2} \text{ RTTs.}$$

ii. **Summing the packets sent over RTT**

During an RTT k , TCP sends approximately $W(k)$ packets. Total packets in the cycle:

$$N = \sum_{k=0}^{K-1} (W_{\min} + k).$$

We can split this into two sums:

$$N = \sum_{k=0}^{K-1} W_{\min} + \sum_{k=0}^{K-1} k = KW_{\min} + \frac{K(K-1)}{2}.$$

And substitute

$$W_{\min} = \frac{W_{\max}}{2}, \quad K = \frac{W_{\max}}{2}.$$

To obtain these terms:

$$KW_{\min} = \left(\frac{W_{\max}}{2}\right) \left(\frac{W_{\max}}{2}\right) = \frac{W_{\max}^2}{4},$$

$$\frac{K(K-1)}{2} = \frac{1}{2} \left(\frac{W_{\max}}{2}\right) \left(\frac{W_{\max}}{2} - 1\right) = \frac{W_{\max}^2}{8} - \frac{W_{\max}}{4}$$

By adding these terms together, we get:

$$N = \frac{W_{\max}^2}{4} + \frac{W_{\max}^2}{8} - \frac{W_{\max}}{4} = \frac{3W_{\max}^2}{8} - \frac{W_{\max}}{4} \approx \frac{3}{8}W_{\max}^2.$$

Thus, the total packets sent through a cycle are:

$$N_{\text{packets}} \approx \frac{3}{8}W_{\max}^2.$$

3 pts. **CWND and Throughput Modeling.** Using the term derived above, show that the steady-state relationship between average CWND $\bar{W}$, loss probability p , and throughput is:

$$T \approx \frac{K \cdot \text{MSS}}{\text{RTT} \sqrt{p}}$$

and compute constant K . State all assumptions.

i. **Average CWND $\bar{W}$**

Assuming steady-state behaviour, we know that within one AIMD cycle, CWND increases linearly from $W_{\min}$ to $W_{\max}$. So, the average window over the cycle is:

$$\bar{W} = \frac{W_{\min} + W_{\max}}{2} = \frac{\frac{W_{\max}}{2} + W_{\max}}{2} = \frac{3}{4}W_{\max}.$$

ii. Relating $W_{\max}$ to loss probability p and solving for constant K

Assume that there is steady-state packet loss, where at least 1 packet is lost per cycle. Then, we can approximate probability p to be defined as:

$$p = \frac{\text{losses per cycle}}{\text{packets sent per cycle}} = \frac{1}{N_{\text{packets}}}.$$

Using the term derived in question 1, we know that

$$N_{\text{packets}} \approx \frac{3}{8} W_{\max}^2.$$

Therefore:

$$p \approx \frac{1}{\frac{3}{8} W_{\max}^2} = \frac{8}{3 W_{\max}^2}.$$

Solving for $W_{\max}$, we obtain:

$$W_{\max} = \sqrt{\frac{8}{3p}}.$$

Which we can use to express $\bar{W}$ in terms of p , obtaining:

$$\bar{W} = \frac{3}{4} \sqrt{\frac{8}{3p}} = \frac{3}{4} \cdot \frac{2\sqrt{2}}{\sqrt{3}} \cdot \frac{1}{\sqrt{p}} = \frac{3}{2} \sqrt{\frac{2}{3}} \cdot \frac{1}{\sqrt{p}}.$$

Therefore allowing us to solve for K ,

$$\bar{W} = K \cdot \frac{1}{\sqrt{p}}, \quad \text{where} \quad K = \frac{3}{2} \sqrt{\frac{2}{3}} \approx 1.2247.$$

iii. Throughput Modeling

Assuming that throughput T is defined as the average CWND divided by RTT, we can express this as:

$$T \approx \frac{\bar{W} \cdot \text{MSS}}{\text{RTT}}.$$

By substituting the relationship between CWND and loss probability found for $\bar{W}$ above and the value of K , we obtain:

$$\begin{aligned} T &\approx \frac{K \cdot \text{MSS}}{\text{RTT} \sqrt{p}} \\ &\approx \frac{1.22 \cdot \text{MSS}}{\text{RTT} \sqrt{p}} \quad \text{with} \quad K = \frac{3}{2} \sqrt{\frac{2}{3}} \approx 1.22. \end{aligned}$$

- 2 pts. **Queueing Delay and RTT.** Considering the serialization delay, derive maximum one-way queueing delay for buffer B and link capacity C . Express measured RTT as a function of RTT_0 , queue occupancy Q , MSS, and C . Specify any additional queueing assumptions if you make any.

i. Deriving maximum one-way queueing delay.

Assume the buffer is FIFO and measured in packets of size MSS bytes. The serialization delay for transmitting one packet on a link of capacity C in bits per second is:

$$\tau = \frac{\text{MSS} \cdot 8}{C}.$$

If the queue is full and contains B packets, then the maximum one-way queueing delay is:

$$T_{\text{max queue}} = B \cdot \tau = B \cdot \frac{8 \cdot \text{MSS}}{C}.$$

ii. Measured RTT as a function of Q , RTT_0 , MSS, and C .

Let Q be the current number of packets in the queue (where $0 \leq Q \leq B$). The corresponding queueing delay is:

$$T_{\text{queue}}(Q) = Q \cdot \tau = Q \cdot \frac{8 \cdot \text{MSS}}{C}.$$

Round-trip time is measured as $\text{RTT} = T_{\text{prop}} + T_{\text{queue}}$. For RTT_0 , assume there is no queueing yet.

Thus, the measured RTT is:

$$\text{RTT}(Q) = \text{RTT}_0 + \frac{8Q \cdot \text{MSS}}{C}.$$

- 2 pts. **Effect of Buffer Size on Web Fetch Times.** Using previous results, explain algebraically why increasing buffer B increases latency for short transfers even when long-term throughput stays high.

As the buffer size B grows, so does the maximum congestion window, W_{max} . By question 2, we know the loss probability is

$$p \approx \frac{8}{3W_{\text{max}}^2}.$$

So, as B (and therefore W_{max}) increases, p decreases. This means packet loss happens less frequently while throughput stays high. However, a bigger buffer means that the queue can hold more packets. By question 3, the RTT is

$$\text{RTT}(Q) = \text{RTT}_0 + \frac{8Q \cdot \text{MSS}}{C}.$$

Since Q can grow larger when B is larger, the RTT increases even though throughput does not drop. This means that a larger buffer helps maintain high throughput, but it also increases latency for shorter transfers because a filled-up queue adds extra delay.